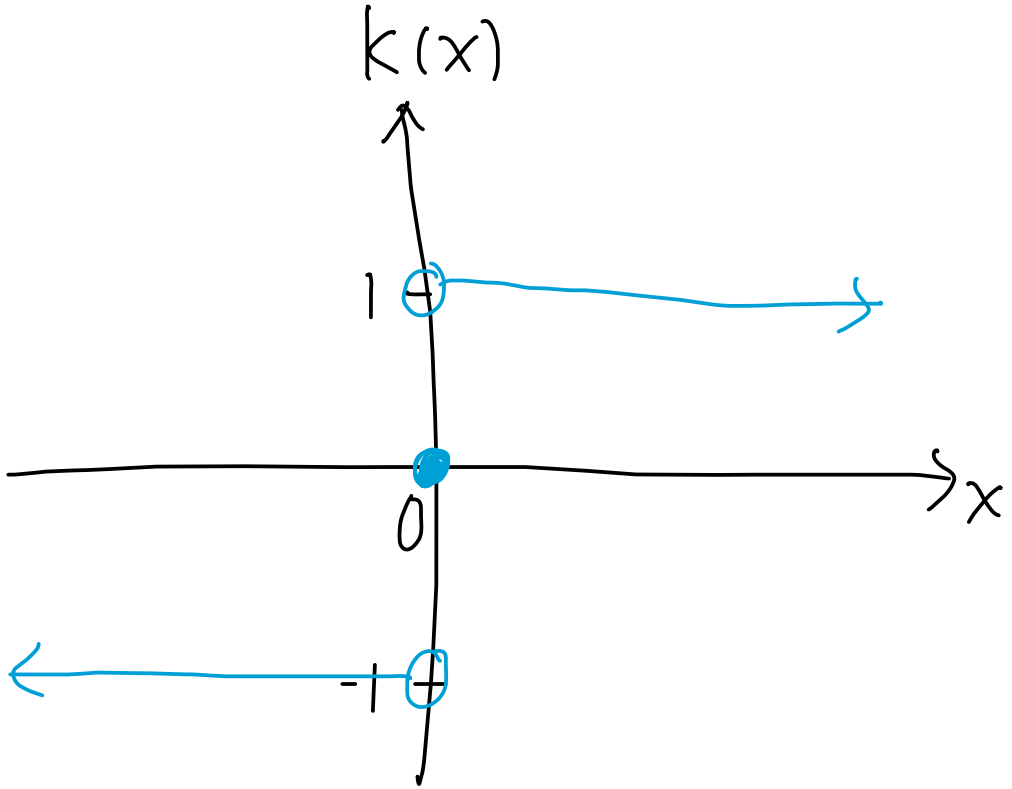
Tutorial Topic 2.

(1) $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} \frac{1}{x+2} = \frac{1}{1+2} = \frac{1}{3}$

$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (1-2x) = 1-2(1) = -1$

$\therefore \lim_{x \to 1} f(x)$ does not exist because $\lim_{x \to 1^-} f(x) \neq \lim_{x \to 1^+} f(x)$

(2)



(a) $\lim_{x \to 0^-} k(x) = -1$

(b) $\lim_{x \to 0^+} k(x) = 1$

(c) $\lim_{x \to 0} k(x)$ does not exist because $\lim_{x \to 0^-} k(x) \neq \lim_{x \to 0^+} k(x)$

(d) $\lim_{x \to 0^-} |k(x)| = \left| \lim_{x \to 0^-} k(x) \right| = |-1| = 1$

(3)

(a) 10

(b) -33

(c) $-\frac{22}{13}$

(d) $-\frac{1}{4}$

(e) $\frac{44}{5}$

(f) 4

(g) -7

(h) -2

(i) $\frac{3}{2}$

(j) 1

(4) $\lim_{x \to -2^-} (x+2) = -2+2 = 0$

$\lim_{x \to -2^+} (2x+4) = 2(-2)+4 = 0$

$\Big\} \lim_{x \to -2} f(x) = 0$

at $x = -2$; $f(x) = x^2 - 4$

$f(-2) = 4 - 4 = 0$

$\therefore f(x)$ is continuous at $x = -2$ because $\lim_{x \to -2} f(x) = f(-2)$

continuity test

(5)(a) $f(x)$ is discontinuous at $x=0$ because $f(0)$ is undefined and $\lim_{x\to 0} f(x)$ does not exist.

(b) $g(x)$ is not continuous at $x=-1$ because $g(-1)$ is undefined.

$$\lim_{x\to -1} g(x) = \lim_{x\to -1} \frac{(x-1)(x+1)}{x+1} = -2 \checkmark$$

(c) $\lim_{x\to 1^-} (x+1) = 2$.

$\lim_{x\to 1^+} (2-x) = 1$

$\Big\}$ $\lim_{x\to 1} h(x)$ does not exist.

$\therefore$ $h(x)$ is not continuous at $x=1$.

(b) Since $f(x)$ is continuous,

$$\lim_{x \to 1} f(x) \text{ exists}$$

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x)$$

$$\lim_{x \to 1^-} (Ax + 5) = \lim_{x \to 1^+} (x^2 - 3x + 4)$$

$$A + 5 = 1 - 3 + 4$$

$$A = -3.$$